Inverse-Design Objectives: Broadband Reflective Polarisation Converter

16x16 SUTD-PRCM unit cell | VAE-latent MOPSO | evaluated by the RDN forward surrogate (no CST in the loop)

Design goal

Under x-polarised normal incidence, reflect the wave as **y-polarised** (cross-pol) with PCR ≥ 0.90 over the widest possible band. The cell is ground-backed, so there is no transmission; conversion requires an anisotropic pattern whose u/v (±45°) reflections have similar magnitude and about 180° phase difference.

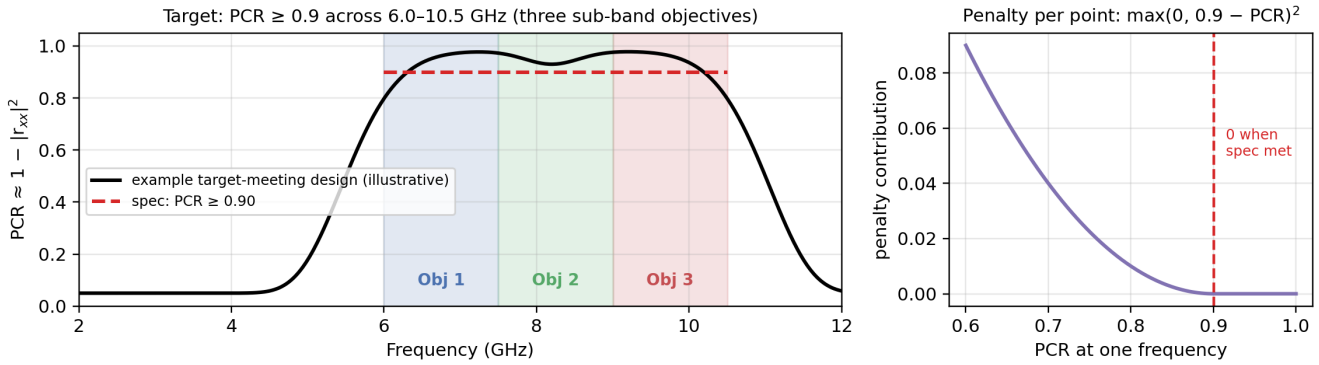
Figure of merit

PCR = $|r_{yx}|^2 / (|r_{yx}|^2 + |r_{xx}|^2)$. The surrogate predicts only complex r_{xx}, so we use **PCR ≈ 1 − $|r_{xx}|^2$** (valid because there is no transmission and absorption is low, so $|r_{xx}|^2 + |r_{yx}|^2 \approx 1$). PCR ≥ 0.9 is equivalent to **$|r_{xx}| \le -10$ dB**. The estimate is most accurate in the high-PCR regime that matters.

Objectives (all minimised)

#	Objective	Band (GHz)	Spec	Penalty
1	PCR, lower sub-band	6.0 - 7.5	PCR ≥ 0.90	$\text{mean}_{\text{band}}\ \max(0, 0.9 - \text{PCR}(f))^2$
2	PCR, middle sub-band	7.5 - 9.0	PCR ≥ 0.90	$\text{mean}_{\text{band}}\ \max(0, 0.9 - \text{PCR}(f))^2$
3	PCR, upper sub-band	9.0 - 10.5	PCR ≥ 0.90	$\text{mean}_{\text{band}}\ \max(0, 0.9 - \text{PCR}(f))^2$

Splitting the band into three objectives exposes the trade-off (designs that cover the low end vs the high end) as a Pareto front and keeps the swarm diverse. A design with all three penalties = 0 is a converter covering 6.0-10.5 GHz: fractional bandwidth ≈ **55 %** at PCR > 90 %. Bands are starting values, to be tuned to where RDN-type designs can actually convert (use the per-objective convergence plot). Each band spans ≥ 10 points of the 0.1 GHz surrogate grid.



Left: spec and sub-bands with an illustrative (not simulated) target-meeting PCR curve. Right: penalty per frequency point; zero once PCR ≥ 0.9.

Success criteria and reporting

• Report the **worst-case PCR per sub-band** (`min_PCR_*` columns in `pareto_summary.csv`) and the continuous band where PCR ≥ 0.9, not the penalty values.

• Strict pass: every point in 6.0-10.5 GHz has PCR ≥ 0.9 (sort by `min_PCR` or set `GOOD_TOL = 1e-6`; the default 1e-3 allows dips to about 0.87).

Assumptions and verification

• **Absorption can mimic conversion:** a lossy resonance lowers $|r_{xx}|$ without producing r_{yx}. The startup sanity check measures absorption on real SUTD data.

• **The optimiser exploits surrogate error:** it seeks where predictions look best, and narrow dips between 0.1 GHz points can be missed.

• **Verification:** simulate the top 5-10 Pareto designs (`pareto_patterns.npy`) full-wave with `cst_interface.py` (both channels, 1001 points) and report predicted vs simulated PCR side by side.